

On-Chip Meta-Optics for Engineering Arbitrary Trajectories with Longitudinal Polarization Variation

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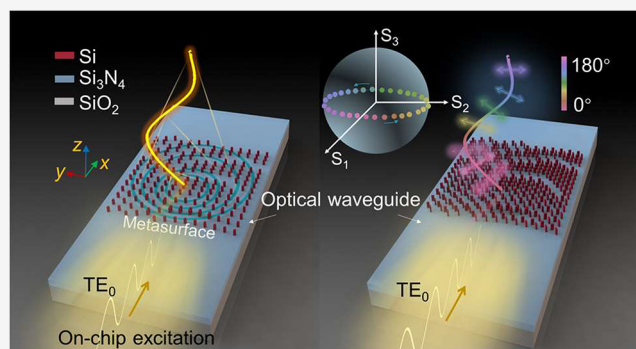
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Supporting Information

ABSTRACT: On-chip integrated meta-optics promise to achieve high-performance and compact integrated photonic devices. To arbitrarily engineer the optical trajectory along the propagation path in an on-chip integrated scheme is of significance in fundamental physics and various emerging applications. Here, we experimentally demonstrate an on-chip metasurface integrated on a waveguide to enable predefined arbitrary optical trajectories in the visible regime. By transformation of the transverse phase to generate longitudinal mapping, the guided waves are extracted and molded into any different optical trajectories (parabola, hyperbola, and cosine). More intriguingly, predefined polarization states with longitudinal variation are also successfully imparted along the trajectory. Owing to the on-chip propagation scheme, the trajectories are uniquely free from zero-order diffraction interference, naturally having a higher signal-to-noise ratio beyond conventional free-space forms. Overall, such on-chip optical trajectory engineering allows for miniaturized integration and can find paths in potential applications of complex optical manipulation, advanced laser fabrication, and microscopic imaging.

KEYWORDS: On-chip meta-optics, Optical trajectory engineering, Optical manipulation, Longitudinal polarization variation



With extensive application potential in various fields including microfabrication, optical tweezers, light bullets, and medical imaging, etc., research on accelerating optical trajectories has attracted considerable interest.^{1–3} Bessel-like beams, as a novel class of accelerating optical beams, propagate along predefined arbitrary trajectories and possess three fascinating properties, namely, self-acceleration, diffraction-free propagation, and self-healing, revealing an emerging research direction.^{4,5} Typically, conventional optical setups, for example, the spatial light modulators for generating Bessel-like beams,^{6–10} are bulky and low diffraction efficiency, and lack flexible trajectory control capabilities, which makes it challenging for the miniature chip integration and hinders their further applications. The emergence of metasurfaces with ultrathin flat structures alleviates this issue,^{11–17} and provides a convenient way to generate accelerating beams by modulating free-space light.^{18–23}

Recently, more advanced studies on metasurfaces integration onto optical waveguides could enable manipulating the guided waves,²⁴ and achieve various functionalities and applications, including on-chip metalens,^{25–27} multiplexed meta-holography,^{28–32} vortex beam generator,^{33,34} augmented reality,^{35–37} and screen display,³⁸ thus providing a new route for miniature on-chip systems and integrated photonic devices. In particular, due to the on-chip propagation scheme, the demonstrated display and imaging functionality from on-chip metasurfaces are uniquely free from zero-order background interfer-

ence.^{35,36,38} Despite the above endeavors, a novel on-chip approach to engineering arbitrary optical trajectories has not yet been achieved and remains to be explored. Furthermore, the on-chip exploit of manipulating the optical parameters along the trajectories is also of great promise, but it remains unveiled due to the lack of encoding degree of freedom and schemes. (Recent research advances are listed in Table S1, Supporting Information.)

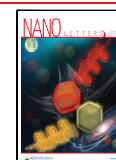
Here, we propose and experimentally demonstrate a platform of on-chip metasurfaces integrated on a waveguide to enable predefined arbitrary trajectories and longitudinally varied polarization in the visible regime. The nanoscale meta-atom array is elaborately patterned onto the optical waveguide and serves to extract the propagating guided waves to generate arbitrary optical trajectories (parabola, hyperbola, and cosine) with a high signal-to-noise ratio beyond the conventional free-space forms. Through further employing the on-chip meta-diatom composed of two orthogonal nanorods, on-chip metasurfaces can achieve simultaneous and independent

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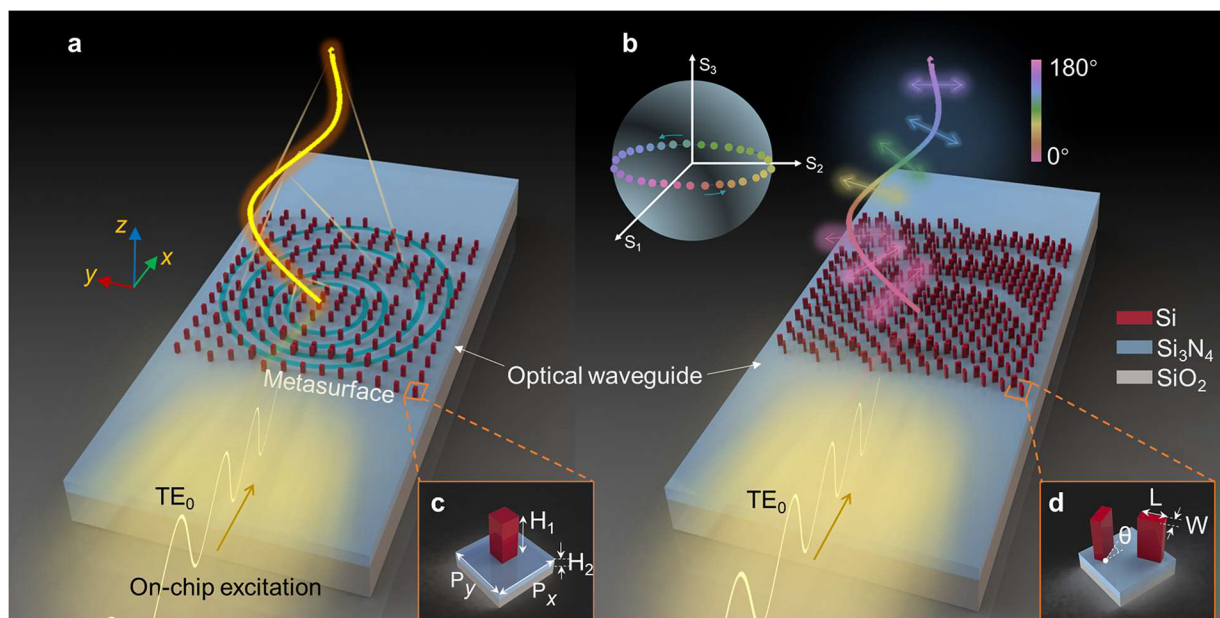


Figure 1. Schematics of on-chip metasurfaces for engineering an arbitrary predefined optical trajectory, and with longitudinal polarization rotation. (a) The on-chip detour phase-only metasurface is integrated onto the waveguide to generate a predefined beam trajectory. (b) By integrating the on-chip meta-diatom array above the waveguide, a Bessel-like beam with the same optical trajectory as in part a but whose polarization rotates along the propagation path is generated. The colorful arrows depict the linear polarization angles at the point of maximum intensity along the trajectory at each z -plane. Perspective view of the unit cell of (c) the on-chip detour phase-only metasurface composed of monatomic α -Si nanoblock and (d) the on-chip diatomic metasurface composed of two orthogonal α -Si nanoblocks. The distance-dependent polarization state of the trajectory varies around the equator of the Poincaré Sphere.

wavefront manipulation and multiple polarization control of the out-coupling lightwave, as a proof of concept, enabling on-chip optical trajectory with polarization state variation along the propagation path. We envision that the on-chip miniaturized meta-optic platform for engineering arbitrary optical trajectories with longitudinal polarization variation has great potential for practical applications in complex optical manipulation, advanced laser fabrication, microscopic imaging, biophysics, and material processing.

Figure 1(a) schematically illustrates the optical trajectory generation of the proposed on-chip metasurface integrated on the waveguide configuration. Specifically, guided waves propagating along the x -direction are extracted by the on-chip detour phase-only metasurface array to generate an arbitrary predefined trajectory. Any point on the trajectory is the apex of a conical ray bundle emanating from a circle on the metasurface plane and interfering to create a transverse Bessel-like field pattern along the predefined propagation path.⁵ The transverse field profile can be described as the Bessel function $J_0(\alpha r)$ (α is the divergent factor of the Bessel-like beam and r is the radius under the polar coordinate system), and it remains invariant main lobe along the predefined trajectory. Furthermore, by the integration of the on-chip meta-diatom array above the waveguide, a Bessel-like beam with the same trajectory as in Figure 1(a) but whose polarization rotates along the propagation path is generated, as shown in Figure 1(b). The distance-dependent polarization state of the extracted guided waves could vary around the equator of the Poincaré sphere.

The proposed on-chip metasurface array consists of amorphous silicon (α -Si) meta-atoms (Figure 1(c)) or meta-diatoms (Figure 1(d)) on top of a planar waveguide of Si_3N_4 ($H_2 = 220$ nm) with a high effective index ($n \sim 2.05$) with a thick (~ 500 nm) SiO_2 substrate underneath. Other alternative

high-index dielectric materials have also been explored and investigated for the metasurface design,^{39–41} and more details can be found in Section S3, Supporting Information. Here, the width W of a single meta-atom is 90 nm, and the length L and the width W of the meta-diatoms are 140 and 70 nm, respectively. The nanostructure height H_1 is 380 nm, and θ is the rotation angle of meta-diatoms. The period P_x/P_y determines the wavelength of the propagating guided waves along x/y -directions.

The design flow of on-chip metasurfaces for engineering the predefined arbitrary trajectories and polarization states varying along the propagation path is shown in Figure 2(a). The phase profiles supposed on the on-chip metasurface consist of φ_1 and φ_2 , where phase φ_1 is utilized to generate a Bessel-like beam with a predefined trajectory, and the function of φ_2 is to modulate the polarization state of the extracted guided wave. A common approach (first proposed by Chremmos and Zhao et al.) to generate accelerating Bessel-like beams is based on the conical ray bundles emitted from circles on the input phase plane.^{4,5} To obtain these conical bundles of rays, we manipulate the extracted guided waves by the on-chip detour phase from the meta-atom displacements, directing the rays to mold an arbitrary optical trajectory. Specifically, we calculate the required phase profile of the predefined trajectory (more details can be found in Section S4, Supporting Information), thus obtaining the positions of Δx_m to encode the detour phase by $\varphi_m(x) = 2\pi \cdot \Delta x_m / P_x$. Here, Δx_m represents the m th meta-atom displacement along the x -direction within a period of P_x . More detailed discussions on the working mechanism of the detour phase can be found in our earlier works.^{36,38} Since the single meta-atom (Figure 1(c)) can merely modulate the phase of the extracted guided waves, to achieve varying polarization manipulation along the beam trajectory, a new design strategy

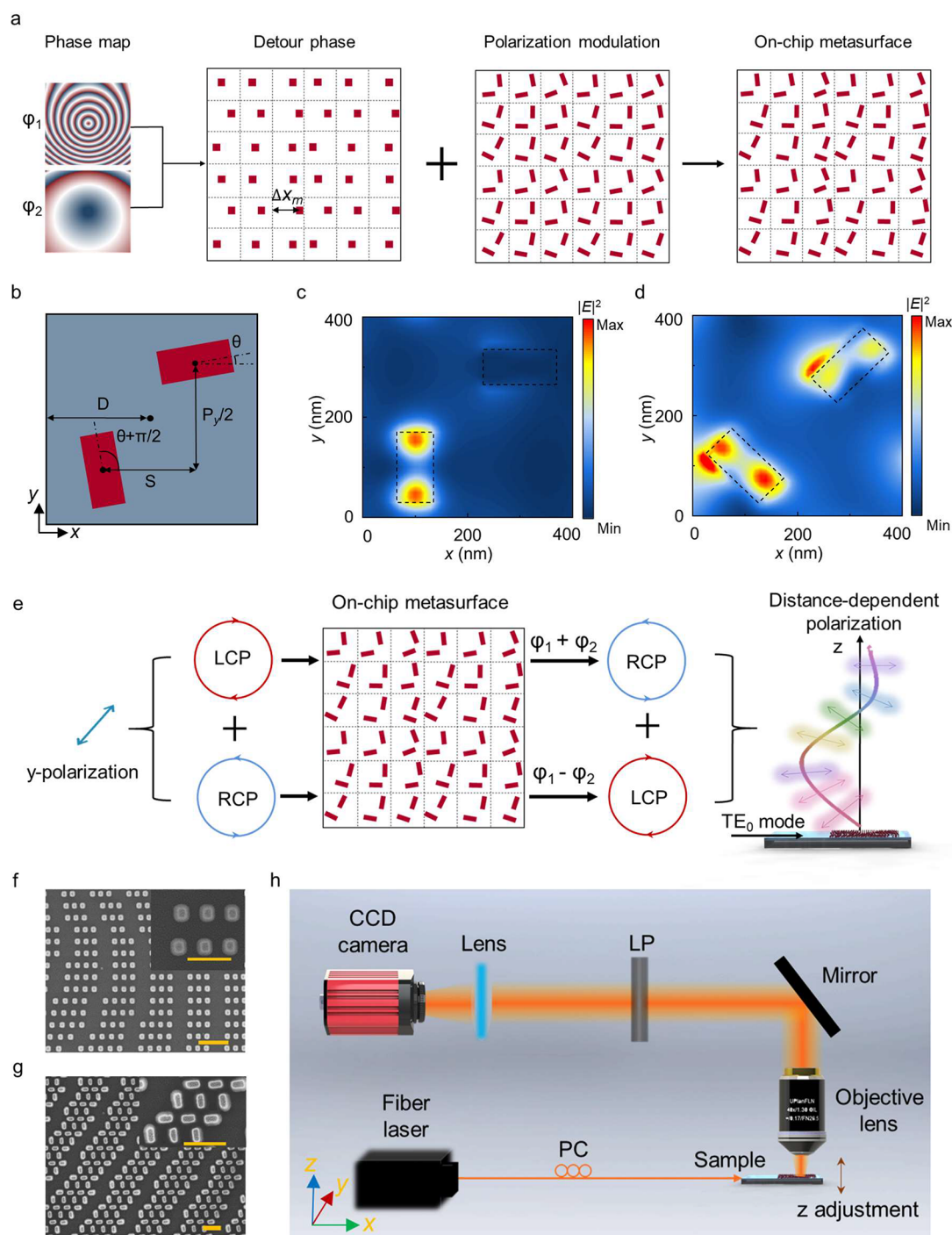


Figure 2. Design of the on-chip metasurface for engineering the optical trajectory. (a) The on-chip metasurface array design method to generate a diffraction-free Bessel-like beam with a predefined trajectory and distance-dependent polarization rotation. (b) Schematic top-view of the on-chip diatomic metasurface pixel composed of two orthogonal nanorods. The corresponding geometric parameters are $P_x = P_y = 400$ nm, $S = 200$ nm, $L = 140$ nm, and $W = 70$ nm. Here, θ is the orientation angle of the meta-atom, and D and S are the global displacement of meta-diatoms and the local displacement between diatomic nanoblocks along x -direction, respectively. (c, d) Electric-field intensity distributions ($|E|^2$) in the xy -plane at $z = 110$ nm when the TE_0 modes incidence from x -directions at the wavelengths of 600 nm, respectively. (e) Principle of generating longitudinally polarized optical trajectories. (f, g) The SEM images of the fabricated on-chip metasurface sample, scale bar: 500 nm. (h) Schematic view of the experimental setup for characterizing the predefined trajectories and longitudinal polarization rotation. PC: polarization controller; LP: linear polarizer.

for on-chip nanostructures with flexible and multiple polarization control capability is required.

To enable on-chip full phase and polarization control, we replace the single meta-atom with the orthogonally aligned meta-diatoms in the unit cell, as shown in Figure 1(d) and

Figure 2(b), where D is the global displacement of meta-diatoms, and S is the local displacement between diatomic nanoblocks along the x -direction. Each nanoblock can be viewed as a dipole antenna that extracts light polarized along the major axis of the nanoblock. The simulated in-plane

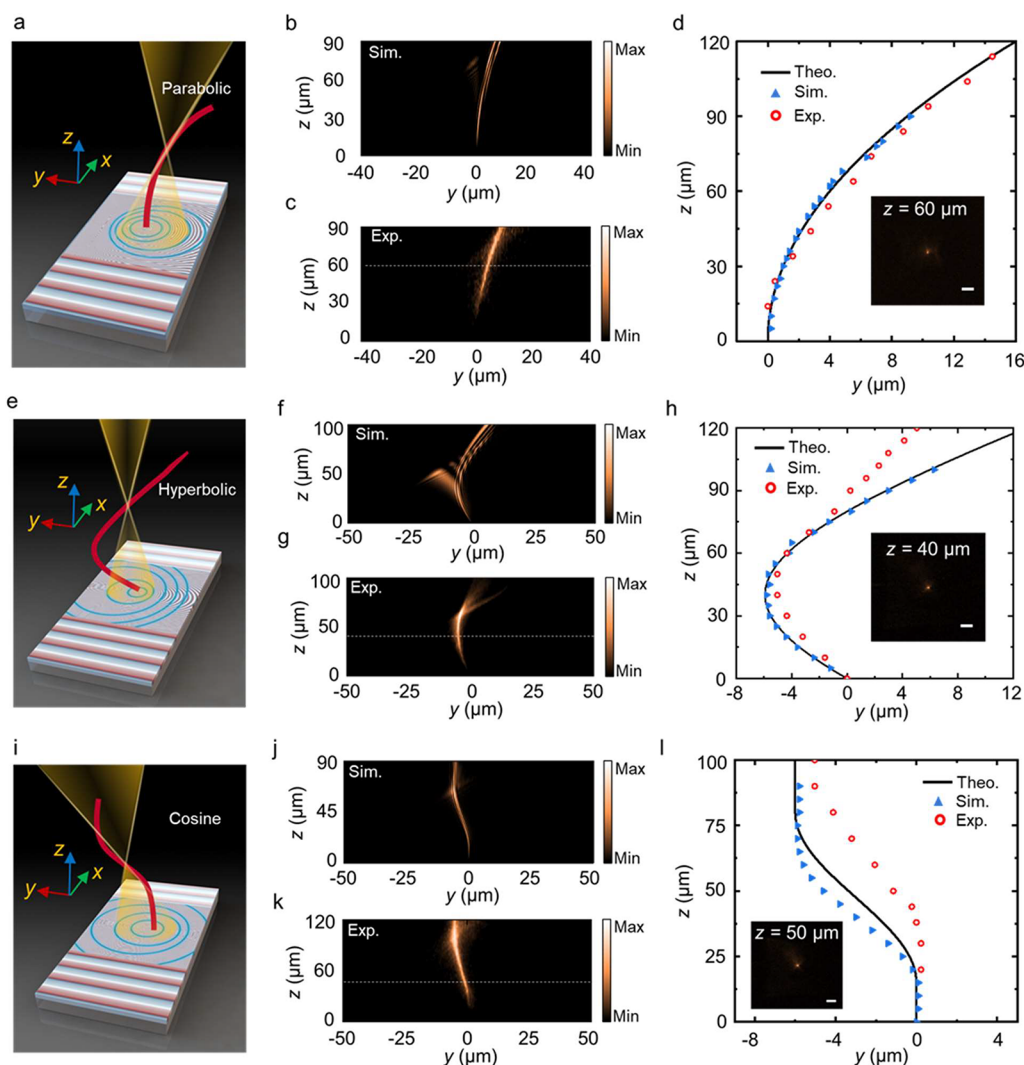


Figure 3. Characterizations of different accelerating Bessel-like beams with predefined arbitrary trajectories. Schematics of the on-chip metasurface utilized to generate (a) parabolic, (e) hyperbolic, and (i) cosine trajectories. Trajectories are (a) a planar parabolic $y = z^2/900$, (e) a planar hyperbolic $y = (3z^2/16 - 15z + 800)^{1/2} - 800^{1/2}$, (i) a planar cosine $y = 0$ ($0 \leq z \leq 15 \mu\text{m}$), $y = 3\cos(\pi(z - 15)/65) - 3$ ($15 \mu\text{m} \leq z \leq 80 \mu\text{m}$), and $y = -6$ ($z \geq 80 \mu\text{m}$). Simulated and experimental longitudinal intensity distributions of the (b, c) parabolic, (f, g) hyperbolic, and (j, k) cosine optical trajectories. (d, h, l) Theoretical (black line), simulated (blue triangle), and experimental (red circle) transverse displacements (x -axis) versus propagation distance (y -axis). The insets in parts d, h, and l are the corresponding transverse intensity profiles extracted from the white dashed lines in parts c, g, and k). Scale bar: $15 \mu\text{m}$.

electric field intensity $|E|^2$ profiles of meta-diatoms with the orientation angle $\theta = 0$ and $\theta = \pi/4$ are shown in Figures 2(c) and 2(d), respectively, which exhibit that the meta-atom with strong anisotropy merely responds to the lightwave with polarization along its longitudinal axis.

When $S = P_x/2$, the propagation phase differences of the extracted light between the meta-diatoms are $\Delta\varphi = 2\pi \cdot S/P_x = \pi$, which means that the phase differences between the two orthogonal field components parallel to each of the two nanoblocks are also π . To avoid the influence of near-field coupling and overlapping at certain local displacements, the displacement between the two orthogonally aligned nanoblocks along the y -direction is designed to be $P_y/2$. Therefore, each unit cell of the on-chip meta-diatomic metasurface performs as a half-wave plate, and different orientation angles θ of the meta-diatoms means rotating the half-wave plate with an angle θ , which will rotate the linear polarization of the extracted guided wave by 2θ . For instance of TE_0 mode

propagating along x -direction, it can be regarded as an in-plane y -linearly polarized wave, and arbitrary linear polarization state conversion can be realized by simply rotating the half-wave plate. As a proof of concept, we simulated the polarized beam-steering performance by extracting the guided waves into the free space at the same deflection angle but with orthogonal linear polarizations (more details can be found in Section S5, Supporting Information). Therefore, it is feasible to engineer the polarization state of the extracted guided waves by tailoring the rotation angle of meta-diatoms.

In addition to the versatile polarization control by the orientation angle, the overall phase modulation can be simultaneously and independently achieved by tailoring the global displacement D . Here, the meta-diatom displacement D can be designed with $D = \varphi_1 \cdot P_x/2\pi$ to provide the required phase shifts of the predefined beam trajectory. Intriguingly, by employing the transverse-to-longitudinal mapping, the modulation of the polarization on the ring of radius r will

correspondingly change the polarization of beams on the propagation path along z -direction. To enable the polarization state rotation along the propagation path, we add an additional distance-dependent polarization control term φ_2 by tailoring the meta-diatoms with corresponding orientation angles on each circle in the transverse plane. Since the in-plane y -linearly polarized waves can be decomposed into left-handed circularly polarized (LCP) and right-handed circularly polarized (RCP) components (Figure 2(e)), Pancharatnam–Berry (PB) phase can be induced as an extra degree of freedom to manipulate the extracted guided waves by engineering the meta-atom orientation (more details can be found in Section S6, Supporting Information). The overall phase shifts are $\varphi_R = \varphi_1 + \varphi_2$ and $\varphi_L = \varphi_1 - \varphi_2$ for RCP and LCP components of the extracted light, respectively. Eventually, through combining the detour phase modulation φ_1 with the polarization control φ_2 , constituting the on-chip meta-diatomic metasurface can enable the Bessel-like beams with predefined trajectory and polarization varying along the propagation path, as shown in Figure 2(e).

Notably, the above-mentioned meta-diatomic design can subtly and effectively eliminate the unwanted unpolarized extracted light through interference and endow the target optical trajectory with a high signal-to-noise ratio. For $S = P_x/2$, the unmodulated circularly polarized radiations by the meta-diatoms are with a phase difference of π induced by the propagation distance, thus exhibiting destructive interference. On the contrary, for the extracted waves modulated by two nanoblocks, such a phase difference resulting from propagation distance will be compensated for by a geometric phase arising from the difference of $\pi/2$ in rotation angles and shows constructive interference (more details can be referred to Section S8, Supporting Information).

We fabricated the designed on-chip meta-optic configuration by employing plasma-enhanced chemical vapor deposition (PECVD) and conventional electron-beam lithography (EBL) technology. More detailed fabrication processing can be found in Section S1, Supporting Information. Figures 2(f) and 2(g) exhibit SEM images of the fabricated meta-atom and meta-diatom samples, respectively. To characterize the predefined trajectory performance, the optical microscope with a 50 $\times$ objective and a charge-coupled device (CCD) camera is utilized to collect the extracted light intensity distributions and record the positions of the center of the beams along the propagation path, as shown in Figure 2(h). See more details for the measurements in Method Section. Here, we numerically simulated the extraction efficiency of our on-chip pattern array, which is $\sim 9\%$ (more detailed discussions on the extraction efficiency can be found in Section S7, Supporting Information).

First, we start from the generation of arbitrary predefined trajectories to exhibit the full phase modulation capability of the novel on-chip meta-optic platform. Based on the on-chip detour phase-only metasurface (Figure 2(f)), we demonstrate three different predefined trajectories of accelerating Bessel-like beams without carrying polarization modulation in the y - z plane: Figure 3(a) is a parabolic $y = z^2/900$, Figure 3(e) is a hyperbolic $y = (3z^2/16 - 15z + 800)^{1/2} - 800^{1/2}$, Figure 3(i) is a cosine $y = 0$ ($0 \leq z \leq 15 \mu\text{m}$), $y = 3\cos(\pi(z - 15)/65) - 3$ ($15 \mu\text{m} \leq z \leq 80 \mu\text{m}$), and $y = -6$ ($z \geq 80 \mu\text{m}$). The simulated corresponding optical intensity distributions of light along the propagation path of the three predefined trajectories are shown in Figures 3b, 3f, and 3j, respectively. In the

experiment, by moving the position of the on-chip metasurface mounted on the stage, the light intensity distribution located in different z -planes can be detected. Then, the intensity distributions in the transverse plane located in discrete longitudinal positions are recorded and superimposed to form a 3D volume, and finally, the longitudinal intensity distributions of the beams are extracted from 3D volumes (Figures 3c, 3g, and 3k). More measurement details can be found in Optical Measurement Section (Methods). The experimentally measured propagation trajectories (red circles) are in good agreement with the theoretical (black curves) and simulated trajectories (blue triangles), as shown in Figures 3d, 3h, and 3l. The insets in Figures 3d, 3h, and 3l show the corresponding beam transverse profiles of the three trajectories at $z = 60 \mu\text{m}$, $z = 40 \mu\text{m}$, and $z = 50 \mu\text{m}$, respectively. The cross-sectional intensity profile extracted from the position ($z = 60 \mu\text{m}$, $z = 40 \mu\text{m}$, and $z = 50 \mu\text{m}$) of the white dashed line in Figures 3c, 3g, and 3k can be found in Section S9, Supporting Information, and the corresponding full width at half-maximum (fwhm) of the measured beam pattern is ~ 800 , ~ 950 , and ~ 900 nm, respectively.

Here, there are some deviations between the experimental and simulated trajectories. On the one hand, it might be caused by the material refractive index mismatch problem, the writing field splicing problem, imperfect fabrication processes, etc. On the other hand, in the measurements, the systematic error of the measuring device, including the flatness and measurement accuracy of the XYZ translation stage, and the postprocessing error may be other important reasons for the deviation between the experimental and simulated trajectories. Nevertheless, these problems can be improved and solved by applying more precise fabrication procedures, accurately calibrating the devices, or using multiple measurement methods. Despite the experiment does not match well with the theory, the measured hyperbolic and cosine optical trajectories in Figure 3 can still behave the corresponding predefined trajectory function variation trend. Moreover, more detailed discussions on the discrepancy between the simulated and theoretical trajectories can also be found in Section S10, Supporting Information.

Furthermore, the self-healing property of the optical trajectory was also demonstrated in the simulation (Section S11, Supporting Information). A silver sphere with a radius of 500 nm was placed on the propagation path of the beam, and the diffracted field was reformed immediately after the obstacle. Herein, the trajectory can be theoretically arbitrary, not limited to only the above three type curves. Compared to the generated optical trajectories by conventional free-space metasurfaces, here the integration of metasurface with a waveguide chip can enable the predefined arbitrary trajectories free from zero-order diffraction interference, and more detailed information can be found in Section S12, Supporting Information.

Next, to confirm the full phase modulation and multiple polarization control capability of the on-chip meta-diatomic configuration for engineering trajectories, we first successfully simulated the Bessel-like beam with a straight-line trajectory (Bessel beam) with longitudinal polarization variation. A more detailed discussion is given in Section S13, Supporting Information. Ultimately, as a proof of concept, we experimentally verify the practical Bessel-like beam with a parabolic trajectory with longitudinal polarization rotation performance, as shown in Figure 4. The relation between the

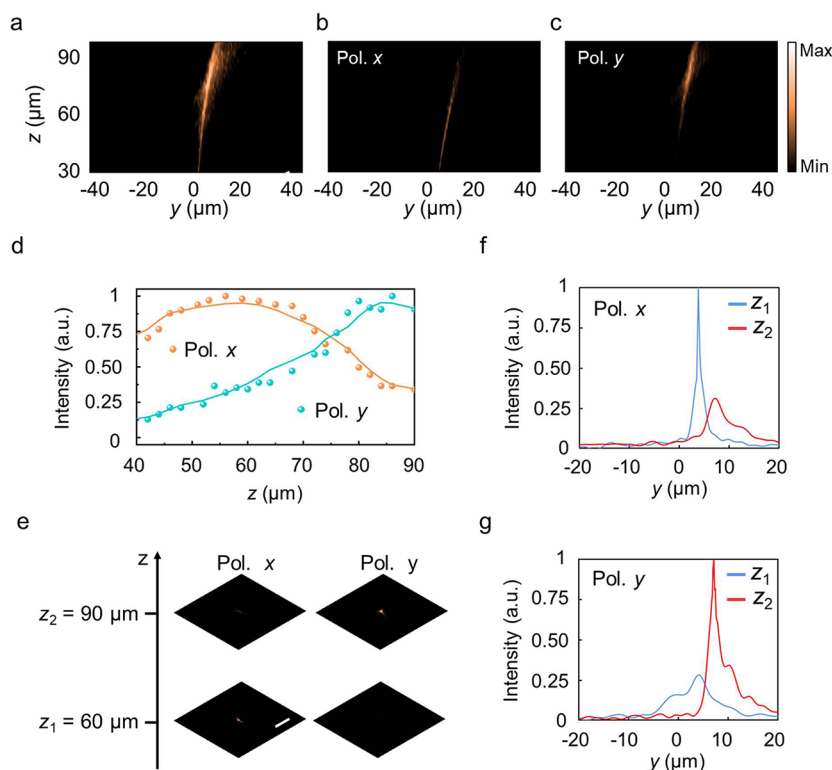


Figure 4. Characterization of the optical trajectory with longitudinal polarization rotation. (a–c) Longitudinal profiles of the generated parabolic Bessel-like beam. (d) Measured normalized trajectory intensity of the *x*–/*y*-polarized component as a function of *z*. (e) Measured transverse profiles of the extracted guided waves at different planes along the *z*-direction. Scale bar: 10 μm . (f, g) Transverse intensity cross-section for the extracted guided waves at different *z*-planes in part e.

polarization rotation angle α and the propagation distance *z* is $\alpha = \pi \times z/100 - 3\pi/10$ ($30 \mu\text{m} \leq z \leq 100 \mu\text{m}$). Figure 4(a) exhibits the *y*–*z* cross-section intensity distributions of the extracted guided waves in the absence of the polarized analyzer, which is basically consistent with that in Figure 3(c). Subsequently, through rotating the polarizer added in front of the CCD for polarization modulation (Figure 2(h)), the *y*–*z* cross-section intensity distributions of the *x*- and *y*-polarized components of the extracted light are obtained, as shown in Figures 4(b) and 4(c). Specifically, the intensity of the *x*-polarized component (Figure 4(b)) varies from strong to weak, while that of the *y*-polarized component (Figure 4(c)) varies in an opposite manner, which suggests that the polarization state of the extracted guided waves changes from *x*-polarization to *y*-polarization as the propagation distance increases.

To more intuitively illustrate the evolution of the polarization state, the normalized intensity of the *x*- and *y*-polarized components are extracted from Figure 4(a) as a function of the propagation distance *z*, as shown in Figure 4(d). The maximum extinction ratio along the propagation path within the region $z = 40 \mu\text{m}$ to $z = 90 \mu\text{m}$ is about -9 dB . Figure 4(e) presents the measured transverse intensity profiles for *x*- and *y*-polarized components at $z = 60 \mu\text{m}$ and $z = 90 \mu\text{m}$. On the $z = 60 \mu\text{m}$ plane, the trajectory central spot mainly consists of the *x*-polarized component, whereas the *y*-polarized component dominates at $z = 90 \mu\text{m}$ plane. The corresponding transverse intensity cross sections at different *z*-planes in Figure 4(e) are shown in Figures 4(f) and 4(g), and the measured average fwhm is $\sim 1.7 \mu\text{m}$. Due to the existence of fabrication and measurement errors, the measured polarization-dependent

intensity variation trends along the trajectory deviate slightly from the theoretical design. Such an on-chip meta-optic configuration is compatible with current photonic integrated circuit technology and can be readily integrated with on-chip photonic devices, which also will provide a new platform for the scalable application of accelerating optical trajectories.

In summary, we propose and experimentally demonstrate a platform of on-chip metasurfaces integrated on a waveguide to achieve predefined arbitrary optical trajectories and longitudinal polarization variation at the visible frequency range. By elaborately engineering the meta-atoms to transform the transverse phase for longitudinal mapping, the on-chip-guided light waves are extracted to mold arbitrary spatial optical trajectories (parabola, hyperbola, and cosine). Moreover, the obtained optical trajectories enjoy being free from zero-order diffraction interference beyond the spatially based metasurface due to the on-chip propagation scheme, thus having a high signal-to-noise ratio. Through further employing the on-chip meta-diatoms consisting of two orthogonal nanorods, we can simultaneously and independently achieve the wavefront manipulation and polarization control of the out-coupling lightwave. More intriguingly, as a proof of concept, the optical trajectories with predefined polarization states and longitudinal variation are successfully imparted based on such on-chip meta-optic configurations with the merit of miniaturization and multidimensional manipulation can provide a new platform for expanding the application of optical trajectories in complex optical manipulation, advanced laser fabrication, microscopic imaging, and biophysics.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.3c04739>.

Methods of simulation, sample fabrication, and optical characterization; recent research advances in on-chip metasurfaces integrated waveguides; alternative dielectric candidates for metasurface design; design principle of trajectories; polarized beam-steering performance; additional numerical simulation of Bessel-like beam; optical extraction efficiency (PDF)

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Author Contributions

Z.Y.L. and Y.S. conceived the idea; Y.S. and S.W. implemented the simulations; Y.S. and Z.L. fabricated the samples; Y.S. performed the optical measurement; Y.S., C.D., and Z.Y.L. drafted the paper, and all authors discussed the results and revised on the paper; Z.Y.L. supervised the work.

Notes

The authors declare no competing financial interest.

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